1. Objective & Context

The objective of this assignment is to build a Python data pipeline **that prepares a dataset for the forecasting team**. This dataset will enable them to cluster different zip codes in Spain based on meteorological characteristics.

The forecasting team is responsible for predicting electricity consumption for customers 12 months in advance. To improve current predictions, they aim to classify zip codes in Spain into 3 to 5 groups based on meteorological data.

The required dataset includes, for each zip code, power category, and year-month, the number of contracts, the maximum and minimum temperatures, and the average relative humidity. The analysis should be performed only on contracts with *client_type_id* equal to 0. Two identical tables are needed: one for customers with solar panels and another for customers without solar panels, as their consumption behavior may differ significantly.

The final tables should resemble the example provided:

zipcode	year	month	power_category	max_temperature	min_temperature	avg_rel_humidity	n
8013	2024	1	Over 5000 kW	21.3	17.8	0.48	982
8013	2024	2	Over 5000 kW	20.9	19.2	0.08	982
8013	2024	3	Over 5000 kW	24.3	23.8	0.83	982
8013	2024	1	Between 3000 kW and 5000 kW	21.3	17.8	0.48	300
8013	2024	2	Between 3000 kW and 5000 kW	20.9	19.2	0.08	300
8013	2024	3	Between 3000 kW and 5000 kW	24.3	23.8	0.83	300
13734	2024	1	Over 5000 kW	15.3	13.1	0.17	123

Finally, since the forecasting team works directly with the Data Warehouse, the resulting datasets must be stored within the database.

2. Deliverables

Create one or more code files that accomplish the following:

1. **Ingestion:** Gather data from the required datasets:
    a. *contracts_eae*
    b. *meteo_eae*
    c. *zipcode_eae*

    Data can be read from a MySQL database or CSV files (you can choose).

2. **Keep only the zip codes with a significant number of contracts:** Process only zip codes with more than 10 customers. From the *meteo_eae* table, retain only data for zip codes that have more than 10 contracts (in the *contracts_eae* table).

3. **Create the power category classification:** Create a new field that classifies the *power_p1* field from the *contracts_eae* table into three categories:
    a. *Power over 5000 kW.*
    b. *Power between 3000 kW and 5000 kW.*
    c. *Power under 3000 kW*

4. **Filter the client type:** Retain only records where *client_type_id* is equal to 0.

5. **Combine the dataframes:** Join the *meteo_eae* table with the *contracts_eae* table.

6. **Solar indicators table:** For contracts with solar panels, create a table that aggregates data **by year, month, zip code, and power category**. This table should include the **maximum temperature, minimum temperature, and average relative humidity**. This table should have the same structure as the example provided in section 1.

7. **Output:** Insert the resulting table (from step 6) into a MySQL database (you can choose the database name).

8. **Non-Solar customers:** Repeat steps 6 and 7 for contracts *without* solar panels.

For this assignment, consider best practices, overall code coherence, and memory management strategies (if applicable); Dask is not required. **Thoroughly comment on the code to explain each step.**

3. Grading

- Data reading and output writing. **(2 points)**
- Main pipeline operations. **(5 points)**
- Code comments**. (1 point)**
- Application of best practices. **(2 points)**